FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

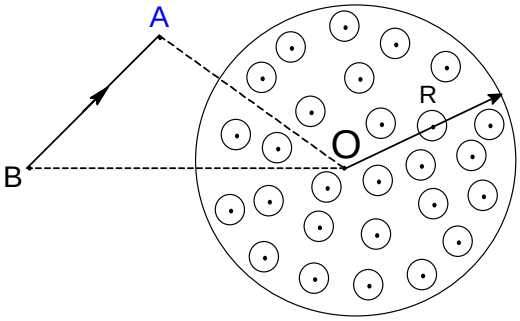
PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Figure shows cross-sectional view of a long solid cylinder of radius 'R'. Current 'I' flows along outward direction along length of cylinder. Then, *find* $\int \vec{B}\cdot\vec{d\ell}$ *along* $\mathrm{B}\vec{\mathrm{A}}$

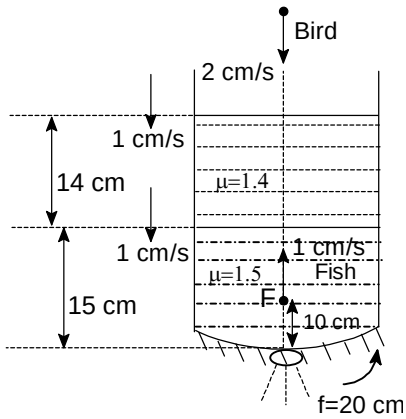
Given: $OB=\left(\sqrt{3}+1\right)R,\ OA=2R,\ \angle OBA=45^{\circ}$. Where 'O' is point of axis of cylinder.



(A) $\mu_0\dfrac{I}{6}$ (B) $-\mu_0\dfrac{I}{6}$

(C) $\dfrac{\mu_0 I\sqrt{7}}{6}$ (D) $\dfrac{-\mu_0 I}{12}$

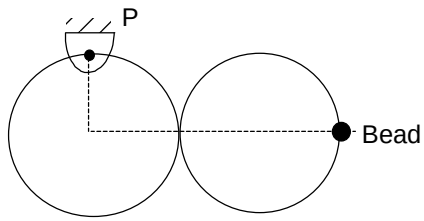
2. A glass cylindrical vessel with base concave mirror of focal length 20 cm is filled with two immiscible liquid of refractive index 1.5 and 1.4. Thickness of liquid level are 15 cm and 14 cm respectively. A fish is moving vertically upward along principle axis of mirror with speed of 1 cm/s. A bird is moving along vertically downward direction with 2 cm/s. A small hole is made at bottom surface of vessel. When distance of fish from pole of mirror is 10 cm, the rate of liquid level falling is 1 cm/s. (focal length of concave mirror is 20 cm). Then, at the given instant



(A) Velocity of image of fish formed after reflection from mirror as seen by bird is 1 cm/s along upward direction.
(B) Velocity of image of fish after formed after reflection of mirror as seen by bird is 1 cm/s along downward direction
(C) Velocity of image of fish as seen by itself is 5 cm/s along downward direction
(D) Velocity of image of fish as seen by itself is 5 cm/s along vertically upward direction

Space for Rough work

3. Two identical ring of equal mass 'm' and radius 'R' rigidly attached wears a bead of mass 'm'. Bead can freely slide without friction. It is hinged at point 'P' in such a way that it can freely rotates about horizontal axis passing through point 'P'. The ring and bead are released from rest from give position. Then, just after release



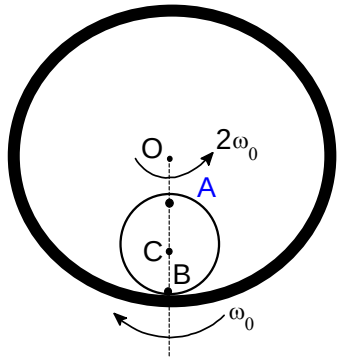
(A) The magnitude of angular acceleration of ring is $\frac{5g}{17R}$

(B) The magnitude of angular acceleration of rod is $\frac{2g}{9R}$

(C) The magnitude of acceleration of bead is $\frac{2g}{7}$

(D) The magnitude of acceleration of bead is $\frac{\sqrt{85}\,g}{9}$

4. A hollow cylinder of radius 4R is rotating about fixed horizontal axis passing through point 'O' with angular velocity ω_0. A solid cylinder of radius 'R' is rolling without slipping with respect to inner surface of hollow cylinder. At the given instant, the line OC has angular velocity of $2\omega_0$. Point A and B are topmost and bottom most point of solid cylinder respectively and C is its centre. Then, at the given instant



(A) Angular velocity of solid cylinder about its own axis is $10\ \omega_0$

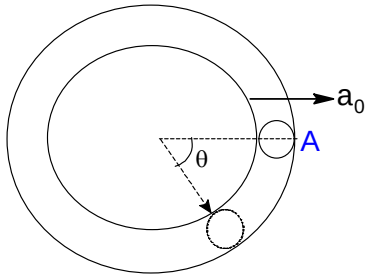
(B) Magnitude of acceleration of point A of solid cylinder is $88\omega_0^{\,2}R$

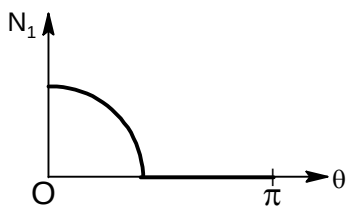
(C) Magnitude of acceleration of point B of solid cylinder with respect to contact surface of hollow cylinder is $112\omega_0^{\,2}R$

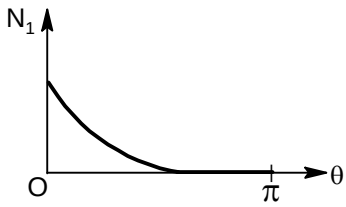
(D) Radius of curvature of point A is $\frac{32R}{11}$

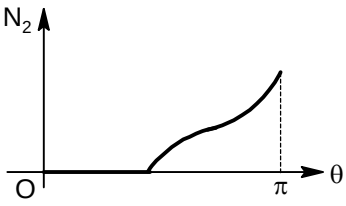
Space for Rough work

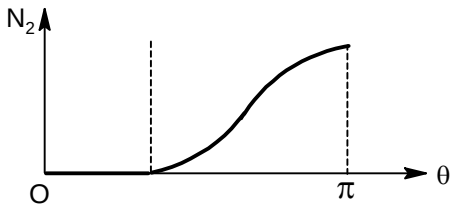
5. A hollow tube containing a ball is kept on smooth horizontal surface. Now, tube is accelerated horizontally and ball is slightly displaced from point 'A'. Let N_1 *and* N_2 be the magnitude of normal force applied by inner and outer surface respectively. Then, which of the following is/are correctly represent approximate nature of N_1 *and* N_2 *Vs* θ $(0 \le \theta \le \pi)$



(A) 

(B) 

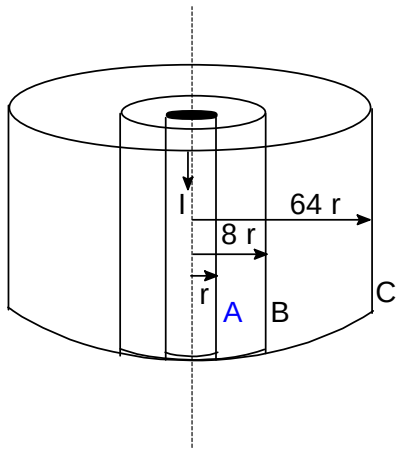
(C) 

(D) 

Space for Rough work

6. A long uniform solid cylinder of radius r and resistivity ρ carries current 'I' along its length which is uniformly distributed throughout its cross sectional area. Two long hollow cylinder B and C of radii 8 r and 64 r surrounds the solid cylinder. The surface of all cylinder A, B and C are perfectly black. At equilibrium, the temperature of B is 400 K.

$\left[\text{Take } \frac{\sigma r^3}{\rho} = \frac{10^{-8}}{\pi^2} \text{S.I unit, } \sigma = \text{Stefan's constant}\right]$. Then, at equilibrium.



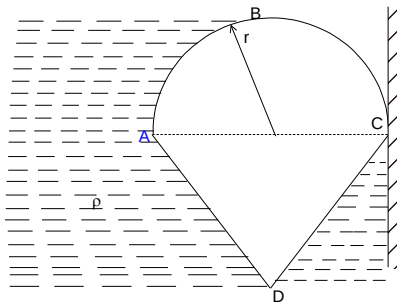
(A) Temperature of A is approximately $565(3)^{\frac{1}{4}} K$

(B) Temperature of C is 200 K

(C) The ratio of temperature of B and C is 2

(D) The current 'I' in solid conductor is $32\sqrt{2}$ A

7. Figure shows cross-sectional view of solid semi cylinder of radius 'r' rigidly attached with solid triangular prism. It floats on a liquid having density 'ρ'. It is in contact with smooth vertical wall. Part BC is not in contact with liquid. The length of object is 'ℓ'. Given (AC = CD = DA). The atmospheric pressure is P_0.



(A) Buoyant force acting on object is independent of atmospheric pressure.

(B) Normal force exerted by vertical wall is $\frac{\rho g r^2 \ell}{2}$

(C) The pressure force exerted by liquid on slant surface ADC of prism is $\left(2+\sqrt{3}\right) r^2 \rho \ell g$

(D) Weight of object is $\left(\sqrt{3}+1+\frac{\pi}{4}\right) r^2 \ell \rho g$

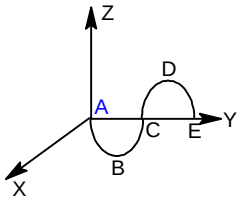
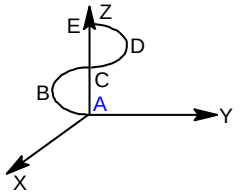
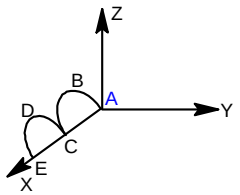
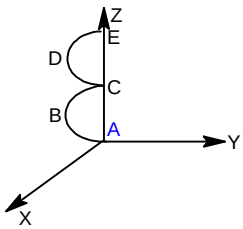
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

In the Column 1, there are conducting loop ABCDE in shape of two semicircular wire of radius 'R'. Column 2 and 3 represent velocity imparted to loop and magnetic field respectively.

	Column – 1		Column – 2		Column – 3
(I)	 Loop lies in xy plane	(i)	$\vec{v} = v\hat{i}$	(P)	$\vec{B} = B\hat{i}$
(II)	 Loop lies in yz plane	(ii)	$\vec{v} = v\hat{j}$	(Q)	$\vec{B} = B\hat{j}$
(III)	 Loop lies in xz plane	(iii)	$\vec{v} = v\hat{k}$	(R)	$\vec{B} = B\hat{k}$
(IV)	 Loop lies in xz plane	(iv)	$\vec{v} = -v\hat{i}$	(S)	$\vec{B} = -B\hat{i}$

8. In which of the following case, $v_E - v_A$ is positive

(A) (I), (i), (R) (B) (II), (ii), (P)

(C) (III), (iii), (Q) (D) (IV), (ii), (S)

Space for Rough work

9. In which of the following case, the magnitude of Induced emf is zero

(A) (I), (i), (R) (B) (II), (ii), (P)

(C) (III), (ii), (P) (D) (IV), (i), (Q)

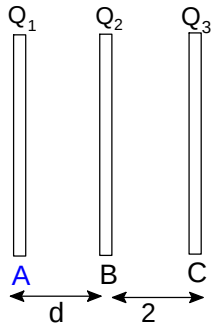
10. In which of the following case, the magnitude of Induced emf is $4vBR$

(A) (I), (i), (Q) (B) (II), (ii), (Q)

(C) (III), (ii), (P) (D) (IV), (ii), (P)

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

Three parallel plates A, B & C of large area carry charge Q_1, Q_2 *and* Q_3 respectively. Values of Q_1, Q_2 *and* Q_3 are listed in Column – 1, Column – 2 and Column – 3 respectively. The separation between A and B is d whereas between B and C is 2d.



	Column – 1		Column – 2		Column – 3
(I)	0	(i)	Q	(P)	0
(II)	Q	(ii)	0	(Q)	-Q
(III)	- Q	(iii)	+2Q	(R)	+2Q
(IV)	2Q	(iv)	-Q	(S)	Q

11. If plate A and C are connected by conducting wire, then in which of the following case the charge flowing through conducting wire is Q/3.

(A) (I), (i), (P) (B) (II), (iii), (R)

(C) (III), (ii), (Q) (D) (IV), (iv), (S)

Space for Rough work

12. If plate A and C are earthed, then in which of the following case, the charge on left surface of plate B is 4Q/3.

(A) (I), (ii), (R) (B) (II), (iii), (Q)

(C) (III), (i), (P) (D) (IV), (ii), (S)

13. If plate B is earthed then, in which of the following cases, the sum of charge of plate A and Plate C is 2Q.

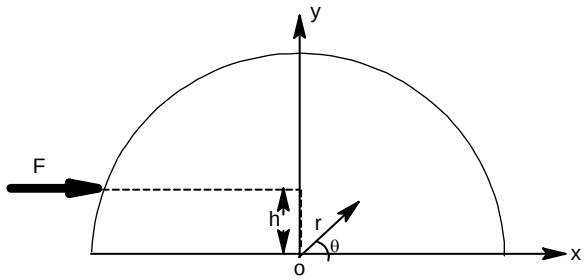
(A) (I), (iv), (R) (B) (II), (iii), (Q)

(C) (III), (iv), (P) (D) (IV), (iv), (S)

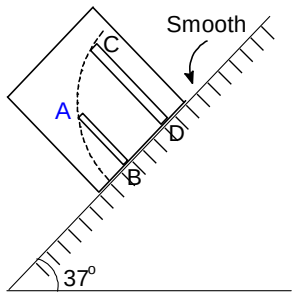
SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. The surface mass density (mass per unit area) of a semi circular disc of radius 'R' varies as $\sigma = \left|\vec{a} \times \vec{r}\right|$, where $\vec{a}$ is a constant velocity along positive x axis and $\vec{r}$ is positive vector with respect to origin. It is kept on smooth horizontal surface with flat surface in contact with surface. Now, force 'F' is applied in such a way that it moves translationally without any rotation. The value of h is $\frac{k\pi R}{16}$ then find value of k.

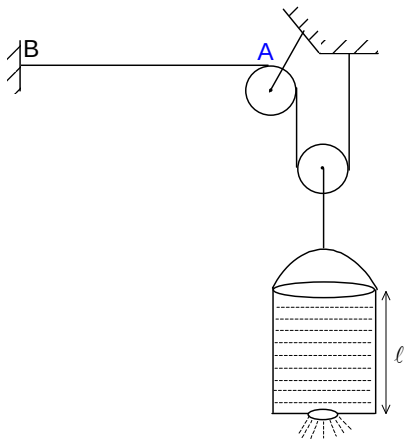


15. Two plate AB and CD are rigidly attached perpendicular to base of a massive box. Box is released on smooth inclined plane having angle of inclination 37°. With what minimum speed with respect to box a particle is projected so as to graze top point A and C of plate $AB = 2m,\ CD = 3m.$ Given Separation between top of plate is 3m. Take $g = 10 m/s^2$.



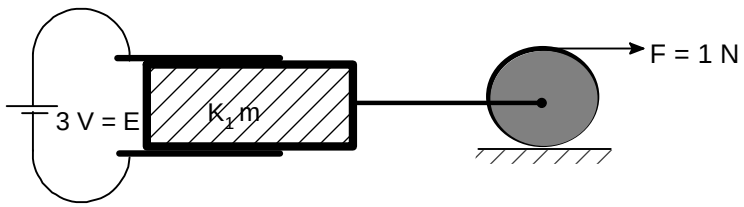
Space for Rough work

16. In the given diagram, there is a cylindrical can of cross sectional area A and length $\ell = 0.4m$ completely filled with liquid of density 'ρ'. It is attached to massless pulley. A small hole of area $'a' = \frac{A}{100}$ is made at the base of can. A wave pulse is generated at point A at an instant when small hole is made at base of can. When can gets empty, the wave pulse reaches point B. Then, find length AB (in meter) $\left(\frac{\rho A}{\mu} = \frac{1}{16} S.I\ unit\right)$ μ is linear mass density of string 'AB' Neglect mass of can and thrust force due to efflux of liquid.



17. There is spherical symmetric charge distribution around origin. The electrostatic potential at a distance 'r' from origin is given as $V = \frac{-K r^4}{4}$ where K is a positive constant. The slope of tangent to density vs 'r' graph at $r = \frac{1}{2} m\ is\ \alpha K \in_0$. Then find value of ''α'.

18. A parallel square plate capacitor of side 'a' is connected to a cell of emf 3 volt. A dielectric of mass $m = 0.1\ kg$ and dielectric constant 'K' initially completely fills the space between plate of parallel plate capacitor. The dielectric is connected to a solid cylinder of mass $m = 0.1\ kg$ through the string. Cylinder is pulled by a tangential force F=1 N and acting on top of cylinder. The surface between cylinder and horizontal surface is enough rough to prevent slipping. The distance between plate is d. The graph between the current through the cell V.S time 't' is a straight line till dielectric leaves the capacitor. Find the slope of straight line $\left[Given \frac{\in_0 a(K-1)}{d} = \frac{1}{9}\right]$.



Space for Rough work

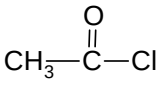
Chemistry

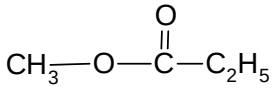
PART – II

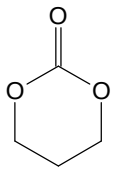
SECTION – A
(One or More than one correct type)

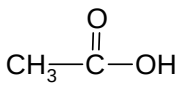
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

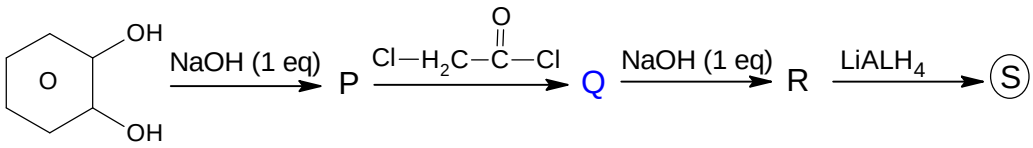
19. Which of the following compound when react with PhMgBr (excess) followed by acidification will yield 3° – alcohol.

(A) 

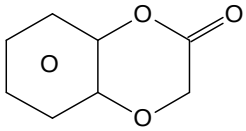
(B) 

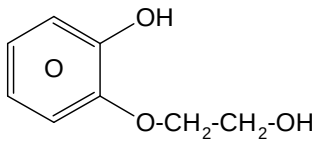
(C) 

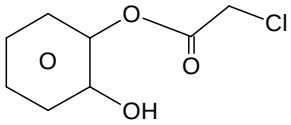
(D) 

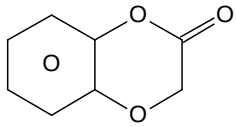
20. 

Select the correct statement

(A) Q is 

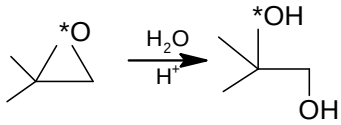
(B) S is 

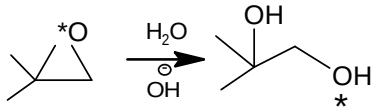
(C) Q is 

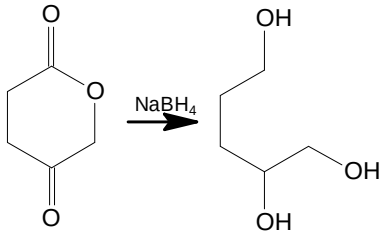
(D) R is 

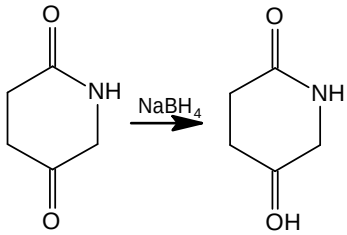
Space for Rough work

21. Select the Incorrect Reaction (s).

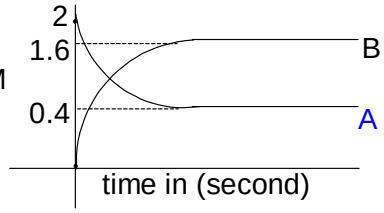
(A) 

(B) 

(C) 

(D) 

22. For the reaction $A \underset{K_2\ \sec^{-1}}{\overset{K_1\ \sec^{-1}}{\rightleftharpoons}} B$



Where is / are correct statement

(A) Equilibrium constant is 4

(B) Time taken for the completion of 50 % of equilibrium concentration of B is 14 sec

(C) Time taken for the completion of 10 % of initial concentration of A is 2.8 sec

(D) Rate constant of backward reaction is $10^{-2}\ \sec^{-1}$

23. Select the correct statement (s) about compound $NOBF_4$

(A) It has $5\sigma\ and\ 2\pi$ bond

(B) N – O bond length in this compound is longer than Nitric Oxide (NO)

(C) It is diamagnetic species

(D) B – F and length in this compound is lower that BF_3

Space for Rough work

24. Assume the complex $\left[Ni\left(PPh_3 \right)_2 \left(SCN \right)_2 \right]$ is paramagnetic

The analogous complex of Pd (II) is diamagnetic

The Isomerism shown by Pd (II) complex is/are

(A) Linkage (B) Optical

(C) Geometrical (D) Polymerization

25. If the radius of First orbit of H - atom is x, which of the following is the correct conclusion

(A) The De Broglie wavelength in the third Bohr orbit of $H - atom = 6\pi x$

(B) The fourth Bohr radius of $He^{+}\, ion = 8x$

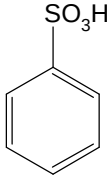
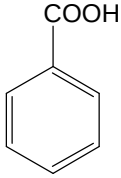
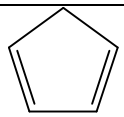
(C) The De Broglie wavelength in third Bohr orbit of $Li\ 2+ = 2\pi x$

(D) The second Bohr radius of $Be^{2+} = x$

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

	Column – 1		Column – 2 (Pka)		Column – 3
(I)		(i)	16	P	Give effervescence with $NaHCO_3$
(II)		(ii)	15.2	Q	React with NaOH
(III)		(iii)	-6.5	R	More Acidic than water
(IV)	CH_3OH	(iv)	4.2	S	Give H_2 gas on reaction with sodium metal

26. Which of the following option is correct.

(A) $I - iii - P$ (B) $II - ii - P$

(C) $III - iv - Q$ (D) $IV - i - P$

Space for Rough work

27. Which of the following option is incorrect.
(A) $I-iii-R$ (B) $II-i-S$
(C) $III-i-S$ (D) $IV-ii-R$

28. Which of the following option is correct.
(A) $I-ii-Q$ (B) $II-iii-R$
(C) $III-i-S$ (D) $IV-ii-P$

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column 1, 2, 3 contains mainly reagent and conditions:

	Column – 1		**Column – 2**		**Column – 3**
(I)	$Conc.\ HNO_3$ / $Conc.\ H_2SO_4$	(i)	$1.O_2;\ 2Dil.\mathrm{H}_2\ SO_4$	P	$1.Mg / \mathrm{Et}_2\mathrm{O}; 2.\mathrm{CO}_2; 3.H_3O^+$
(II)	$HeCl / AlCl_3$	(ii)	$Br_2 / NaOH$	Q	$Conc.\ HNO_3 / Conc.\ H_2SO_4$
(III)	$Ac_2O / AlCl_3$	(iii)	Br_2 / Fe	R	$CO_2 / NaOH;\ H_3O^+$
(IV)	$\mathrm{Pr}opene,\ H_3PO_4$	(iv)	$K_2Cr_2O_7 / Conc.H_2SO_4$	S	H_3O^+

29. Benzene is converted into a mixture of two carboxylic acid using the reagent from the above table in the sequence.
(A) $(II)(iv)(Q)$ (B) $(II)(iii)(P)$
(C) $(III)(ii)(S)$ (D) $(IV)(iv)(R)$

30. Which of the following combination of reagent convert benzene to salicyclic acid.
(A) $(IV)(i)(R)$ (B) $(III)(ii)(S)$
(C) $(II)(iv)(R)$ (D) $(I)(i)(P)$

31. Benzene become benzoic acid on treatment with
(A) $(II)(ii)(R)$ (B) $(III)(iv)(Q)$
(C) $(III)(ii)(S)$ (D) $(IV)(i)(R)$

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. For the equilibrium
$CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4 \cdot 3H_2O + 2H_2O(g)$
$kp = 2.25 \times 10^{-4}\ atm^2$
Vapour pressure of water is 22.8 torr at 298 k the percentage relative humidity is expressed as $x\%$ find $\frac{x}{10}$.

33. The rate constant of a reaction at $27^{\circ}C\ is\ 3\times10^{-3}\ \min^{-1}$. The fraction of molecule crossing the energy barrier if the reaction at that temperature is $10^{-3}\%$. If temperature is increased, the maximum value of rate constant K can be attained is $x\times100\ \min^{-1}$. The value of x is

34. Number of B – O bond in Peroxoborate is x and number of mole of $H_2(g)$ given by hydrolysis of 1 mole of B_2H_6 is y then find $\frac{x+y}{2}$

35. 2 L of a butter solution is prepared that is $1M\ NaH_2PO_4\ and\ 1M\ Na_2HPO_4$. The solution is divided into two halves between two compartment of an of an electrolyte cell using Pt-electrodes. Assume that the only reaction is the electrolysis of water. If the electrolysis is carried out for 212 min with a current of 1.25 A. pH of the anode compartment is approximately
$\left[Pka\left(H_2PO_4^-\right) = 7.1\right]$

36. A gaseous substance A_n remains partially dissociated $A_n(g) \rightleftharpoons nA(g)$. If the given sample of gas has $A_n(g)\ 52\%$ dissociated and the equilibrium mixture diffuses 1.25 times slower than pure oxygen gas under identical condition determine 'n' Atomic weight of A is 32.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Which of the following is (are) correct.
(A) The three co-ordinate plane divide the whole system in 8 octane
(B) The perpendicular distance of the point $(3\cos\theta,\ 3\sin\theta,\ 4)$ from the z axis is constant
(C) In a tetrahedron, each of four vertices is the intersection of four lines
(D) The Y-Z plane divide the line joining the points $(2,\ 4,\ 5),\ (3,\ 5,\ -4)$ in the ratio 2 : 3 (internally)

38. Given the equation of two circles as
$C_1 : z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + Y = 0$ *and* $C_2 : z\bar{z} + \bar{\beta}z + \beta\bar{z} + \delta = 0$ *where* $\alpha,\ \beta \in C$ & $Y,\ \delta \in R$
(A) C_1 & C_2 will touch externally if $|\alpha - \beta| = \sqrt{|\alpha|^2 - Y} + \sqrt{|\beta|^2 - \delta}$
(B) C_1 & C_2 will intersect orthogonally iff $\alpha\bar{\beta} + \bar{\alpha}\beta = Y + \delta$
(C) Radius of circle C_1 is $\sqrt{|\alpha|^2 - Y}$
(D) None of these

39. $I = \int_0^1 \frac{dx}{1 + x^{\frac{\pi}{2}}}$ then
(A) $I > \ell n2$ (B) $I < \ell n2$
(C) $I > \frac{\pi}{4}$ (D) $I < \frac{\pi}{4}$

Space for Rough work

40. The Value (s) of 'a' for which the area of the triangle included between the axes and any tangent to the curve $x^a y = \lambda^a$ is constant is/are

(A) $-\frac{1}{2}$ (B) -1

(C) $\frac{1}{2}$ (D) 1

41. The solution of $P + \cos Px \sin y = \sin Px \cos y \left(P = \frac{dv}{dx} \right)$

(A) $y = 0$ (B) $cx^2 - y = \sin^{-1} x$

(C) $cx - y = \sin^{-1} c$ (D) $y = \sqrt{x^2 - 1} - \sin^{-1} \frac{\sqrt{x^2 - 1}}{x}$

42. Let $h(x) = f(x) - (f(x))^2 + (f(x))^3$ for all real values of x, then incorrect option is

(A) $h(x)$ is increasing wherever $f(x)$ is increasing

(B) $h(x)$ is increasing whenever $f(x)$ is decreasing

(C) $h(x)$ is decreasing whenever $f(x)$ is decreasing

(D) Nothing can be said in general

43. $\lim_{n\to\infty} \left(\frac{3nc_n}{2nc_n} \right)^{\frac{1}{n}} = \frac{a}{b}$ where a & b are relative prime then

(A) $a + b = 43$ (B) $ab = 390$

(C) $a + b = 41$ (D) $ab = 432$

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

	Column – 1 $\left[I(\alpha)\right]b$		**Column – 2** $\left[I(\alpha_1)\right]$		**Column – 3** $\left[I(\alpha_2)\right]$
(I)	$I(\alpha)=\int_0^1 \frac{x^\alpha-1}{\ell n x}dx$	(i)	$\frac{\pi}{2}\ell n\frac{3}{2}$	(P)	$\pi\ell n\left(\frac{\sqrt{5}}{2}\right)$
(II)	$I(\alpha)=\int_0^1 \frac{\tan^{-1}\alpha x}{x\sqrt{1-x^2}}dx$	(ii)	$\ell n\frac{3}{2}$	(Q)	$\frac{\pi}{4}\left(\sqrt{15}-4\right)$
(III)	$I(\alpha)=\int_0^1 \frac{\ell n\left(1-\alpha^2x^2\right)}{x^2\sqrt{1-x^2}}dx$	(iii)	$\frac{\pi}{2}\left(\sqrt{3}-2\right)$	(R)	$\frac{\pi}{2}\ell n\left(\frac{\sqrt{17}+4}{4}\right)$
(IV)	$I(\alpha)=\int_0^\infty \frac{\tan^{-1}\alpha x}{x\left(1+x^2\right)}dx$	(iv)	$\frac{\pi}{2}\ell n\left(\frac{\sqrt{5}+1}{2}\right)$	(S)	$2\ell n\left(\frac{\sqrt{5}}{2}\right)$

44. If $\alpha_1=\frac{1}{2}$ & $\alpha_2=\frac{1}{4}$ then which of the following option is correct?

(A) $I, (i), P$ (B) $I, (ii), S$

(C) $I, (i), S$ (D) $I, (ii), P$

45. If $\alpha_1=\frac{1}{2}$ & $\alpha_2=\frac{1}{4}$ then which of the following option is correct?

(A) $II, (iv), Q$ (B) $II, (iii), R$

(C) $III, (iii), Q$ (D) $III, (iv), Q$

Space for Rough work

46. If $\alpha_1 = \frac{1}{2}$ & $\alpha_2 = \frac{1}{4}$ then which of the following option is correct?

(A) $IV, (i), S$ (B) $IV, (ii), S$

(C) $IV, (i), P$ (D) $IV, (ii), P$

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

$(1-\alpha)x^2 - 2\alpha xy + (1-\alpha)y^2 + (18\alpha - 2)x + (18\alpha - 4)y + (5 - 81\alpha) = 0$ represent a conic section.

	Column –I (α)		Column – II (Conic section)		Column – III (Latus rectum)
(I)	$\frac{1}{8}$	(i)	Parabola	(P)	$2\sqrt{2}$
(II)	$\frac{1}{2}$	(ii)	Ellipse	(Q)	$4\sqrt{2}$
(III)	1	(iii)	Hyperbola	(R)	$8\sqrt{2}$
(IV)	2	(iv)	Rectangular hyperbola	(S)	8

47. Which of the following is correct combination?

(A) $I, (iii), P$ (B) $I, (iv), P$

(C) $II, (i), Q$ (D) $II, (ii), P$

48. Which of the following is correct combination?

(A) $III, (i), S$ (B) $III, (iv), S$

(C) $III, (ii), S$ (D) $III, (iii), S$

49. Which of the following is correct combination?

(A) $I, (ii), P$ (B) $III, (iii), P$

(C) $II, (iii), R$ (D) $IV, (iii), R$

Space for Rough work

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If $y^2\left(y^2-6\right)+x^2-8x+24=0$ and the minimum value of x^2+y^4 is m and maximum value is M, then find the value of $\frac{M}{6m}$

51. Let e the eccentricity of a hyperbola and $f\left(e\right)$ be the eccentricity of its conjugate hyperbola. If n is an even positive integer, then $\int_1^3 \underbrace{fofofo.....of\left(e\right)}_{n\ times}$ is equal to 'a' then 2a equal to

52. $P\left(x\right)=x^5+x^2+1$ *has roots* x_1, x_2, x_3, x_4, x_5 *and* $g\left(x\right)=x^2-2$ *then*
$g\left(x_1\right)\cdot g\left(x_2\right)\cdot g\left(x_3\right)\cdot g\left(x_4\right)\cdot g\left(x_5\right)-30g\left(x_1\cdot x_2\cdot x_3\cdot x_4\cdot x_5\right)=4x+3$
then n equal to

53. The equation of the curve obtained by reflecting the ellipse $\frac{\left(x-4\right)^2}{16}+\frac{\left(y-3\right)^2}{9}=1$ about the line $x-y-2=0$ *is* $16x^2+9y^2+k_1x-36y+k_2=0$ then number of prime factor of $\left(k_1+k_2\right)$ is

54. $\bar{a}, \bar{b}$ & $\bar{c}$ be three non-coplaner vector & $\bar{d}$ be a non zero vector, which is perpendicular to $\left(\bar{a}+\bar{b}+\bar{c}\right)$ *and* $\bar{d}=\sin x\left(\bar{a}\times\bar{b}\right)+\cos y\left(\bar{b}\times\bar{c}\right)+2\left(\bar{c}\times\bar{b}\right)$ then minimum value $x^2+y^2=\frac{a\pi^2}{b}$ *where* a & b are coprime then $a+b$ is

Space for Rough work